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A channel independent generalized seizure detection method for pediatric epileptic seizures

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ABSTRACT

Background and objective: Epilepsy the disorder of the central nervous system has its worldwide presence in roughly 50 million people as estimated by the World Health Organization. Electroencephalogram (EEG) is one of the most common and non-invasive ways of analyzing and studying the subtle changes in neuronal activity of the brain during an epileptic seizure attack. These changes can be analyzed for developing an automated system that would assert the chances of an impending seizure. As changeable nature of seizure affects the patients from having a normal life, hence progress in developing new methods will improve the quality of life and also provide assistance in the medical sector. Objective of the proposed method is to avoid EEG channel selection and use all input EEG channel features to design a generalized epileptic seizure detection framework.

Method: In this work, a long short-term memory network has been proposed that is not complex and has the capability of effectively detecting epileptic seizures from both non-invasive and invasive electroencephalogram recordings. The proposed framework is simple and effective and designed in such capacity that raw electroencephalogram signals can be used to detect seizures. Also, a generalized approach has been followed that is channel independent such that EEG signals belonging to any hemisphere of the brain can be detected effectively by the proposed architecture.

Results: The automated seizure detection system achieved high seizure detection sensitivity of 99.9%, and a low false-positive rate of 0.003 per hour for the Children's Hospital Boston-Massachusetts Institute of Technology dataset. While for the Sleep-Wake-Epilepsy-Center of the University Department of Neurology at the Inselspital Bern dataset, the sensitivity is 99.4% and false-positive rate of 0.006 per hour. Convergence analysis of the proposed model provides a significant amount of reliability and correctness in the efficient detection of epileptic seizures.

Conclusion: Assessment of the proposed framework on non-invasive as well as invasive EEG signals showed that the framework worked well for different type of EEG recordings as different metrics gave satisfactory results. As the framework is simple and did not require any additional parameter optimization techniques, it reduced the processing overheads without affecting the accuracy. Hence, it can be used as an efficient method for monitoring epileptic seizures.

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1. Introduction

A non-communicable and one of the significantly common neurological disorders that affect a considerable number of the world population is epilepsy [1]. Swift and recurring unprovoked seizure is exhibited during an epilepsy attack [2]. This hyper-synchronous firing of neurons in the brain also known as seizure leads to uncontrolled body movement [3]. It is roughly estimated that two-third of patients suffering from epilepsy can be cured or to an extent controlled using available anti-epileptic drugs [4]. Regardless of the availability of medication for treating seizure, a large percentage of patient suffer from recurrent seizures as pathological as well physiological changes related to epilepsy is still not completely deciphered [5]. As epilepsy conditions are heterogenous, understanding the change in dynamics of the brain at the molecular level over the passage of epileptogenesis poses a challenge [10].

Usually, the onset of an epileptic seizure attack is a widespread manifestation of abnormal electrical activity that is presented by the noticeable changes in usual brain activity [6]. Spike as well as slow waveforms [7,8] mark these changes and are considered as the trademark signature of epileptic seizure [19]. To analyze these marked changes, the most extensively accepted non-invasive technique used by physicians for studying brain activities is through

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electroencephalogram (EEG) [8] that provides commendable temporal resolution. Precise epileptic events can be captured using electroencephalogram and different ictal and interictal phases can be easily distinguished from normal events [11]. Expert intervention by neurologists is needed for visual evaluation of the EEG recording. As analyzing an enormous amount of EEG recordings physically is time-consuming as well as exhausting, automatic diagnosis of seizure events is gaining interest [9]. An automated seizure detection system [14,22] would pave way for easier and faster diagnosis of seizure events with minimal misjudgment and error. Although a study dedicated to providing critical facets of preictal duration [20] and its morphology still lacks substantial evidence, yet advancement in developing seizure detection systems would transform patient treatment and management [15,4]. Devising detection strategies is a challenging task as seizures are patient-specific with distinct characteristic features of each seizure [16]. To achieve precise seizure detection, the incorporation of a wide range of temporal and spatial information is necessary [12]. As patients require medication for an extended period, therefore, strategizing an alarm generation prior to seizure onset would be beneficial for quick human intervention [17,18].

The last few decades have seen studies dedicated to efficient seizure detection using different types of algorithms [23]. Automated seizure detection principally consists of two major steps [22]. Analysis and processing of the signals for relevant features constitute the first step. Frequency domain, time domain, time-frequency domain, and non-linear methods have been predominantly used for extracting characteristic features from EEG signals [13]. The second stage consists of classifying the signals using the extracted temporal and spatial features. Traditional classifiers generally adhere to the two stages for automated seizure classification. Primarily two distinguishing approaches are heavily relied upon for the detection of epileptic seizures [24]. In one approach, preictal and interictal recordings are considered to categorize and differentiate between the states while in the second approach, a threshold value is considered for different features which when exceeded leads to the generation of an alarm.

Even though the two-stage method is complex and involves certain command over signal processing but has achieved exceptional result. Advancement in machine learning and breakthrough in deep learning have paved the way for designing models that are capable of learning features automatically. Onset and offset of seizure attack in long EEG recordings have been studied by [25] using a neural network-based model called ScoreNet. The ScoreNet used an adjacent epoch scoring technique to identify seizure epochs before assembling them in a group. Predicted values from the epochs are analyzed for identifying the seizure onset and ending. Also, a different cost function (log-dice loss) [25] that is based on dice similarity has been used by the authors to formulate the loss function of their proposed method. Although variance [26] is commonly used as feature to observe the changes in EEG signals during seizure attack, but there is uneven variation in the signal spectrum for patients suffering from epilepsy. The energy difference during normal event and seizure varies unevenly across sub-bands. Hence in [26], apart from variance, power spectral densities of the EEG signals are also considered for calculating the sub-band energies.

Time-frequency variance has been used by [28] using anomaly generative adversarial network (AnoGAN) to learn the spectral changes in ictal events of EEG signals. AnoGAN is a type of deep convolutional generative adversarial network that has the capability of learning visual information. Wen et al. [27], used autoencoders along with deep CNN for extracting optimal features from EEG signals. Features extracted using autoencoders have the capability of enhancing the speed of convergence and thus lowering the training time of the deep learning model. Sparse denoising autoencoders have been used for learning the features in an unsupervised manner from different EEG channels by [29] for effective identification of brain abnormalities. To avoid preprocessing of EEG data, a CNN with multiple convolutional blocks has been used to learn the relevant features from EEG signals. These extracted features are then used as input for a nested LSTM model to understand and explore the inherent characteristics of EEG signals [30]. Each EEG signal was decomposed using a sliding window into segments with fixed sampling points. The segmentation of the EEG signal was done to assess the impact of each segment on the signal and how it affected the recognition capability of their proposed method.

In [31], authors combined more than one convolutional layer along with max-pooling and fully connected layers to extract features from EEG signals to discriminate between normal, preictal and seizure phases. Tsiouris et al. [32] in their work used various features pertaining to time domain, frequency domain, correlation and graph theory. A 5 sec segment has been used for extracting each feature technique before being used in the LSTM classifier. Different LSTM configurations have been tested in a patient-specific approach to differentiate between preictal and different subgroups of interictal classes. Spectral and temporal features have been extracted using a deep CNN model by Hossain et al. [33], for automated seizure detection and to understand the nuances pertaining to seizure such that the developed model is less prone to variations and changes in the EEG signals. The visualization technique helped in mapping brain activities that would help in locating seizures and also showed the distribution of spatial features across different frequency bands Thara et al. [35], showed in their work that a good scaling technique is equally important and used a CNN model for extracting the features with a merger of increasing and decreasing sequence to point out the characteristics of the signal waveform.

Data augmentation as well as merger of increasing and decreasing sequence has been used by the authors to highlight the information obtained from the samples using perceptual organizational principle [36]. Different merger rules were used for merging different waves and peaks present in the EEG signal. This helped in not only highlighting the shape of the peaks in the signal but also capturing the slight fluctuations. Before using deep learning algorithm, EEG signals have been transformed into gray images using short time Fourier transform [37] as it contained necessary information in time and frequency domain that would be helpful for detecting epileptic seizures. Convolutional neural network has been used to extract features for classifying between preictal and interictal phases by Usman et al. [38]. Authors replaced the fully connected layer of the CNN with a traditional classifier such as support vector machine (SVM) for identifying different states. To preserve the inherent nature of EEG signals, local mean decomposition (LMD) and statistical features have been used in [39]. The Bi-LSTM architecture comprising of two separate LSTM networks propagated from opposite direction and used these decomposed features to identify seizure segments.

Zhou et al. [40], used time and frequency domain features from the preictal, interictal, and ictal segment for distinguishing between seizure and non-seizure signals. The results showed frequency domain signals to edge over time-domain signals in terms of classification accuracy. Authors in [41] represented the EEG signals with reconstructed phase space (RPS), to capture the non-linear dynamics associated with the signal. Transfer learning has been applied to pre-trained convolutional network with the RPS images of EEG signals. The RPS captured the dynamics of EEG signal at a particular time by constructing the state space of the signal. Hence change in signal at any time is reflected by the RPS image. This helped the deep learning network in identifying seizures. Transfer learning has also been used in [42] for designing cross

patient seizure detection CNN model. Authors used deep residual network (ResNet) structure for detecting nocturnal frontal lobe epilepsy. Images of EEG signals have been used by [43] for designing a fully automated CNN model that detected seizures with a false alarm rate of less than 0.5 /h for most of the patients.

Cho and Jang [21], used 2D images with different input modalities for analyzing a combination of network structures that would be suitable for seizure detection. Plot images of long-term EEG recordings have been used by [44], for distinguishing seizure images from non-seizure images. A variant of LSTM also known as deep convolutional LSTM (C-LSTM) has been used for studying brain activity and detecting seizures [45]. While Nagabushanam et al. [46], used a two-layer LSTM model for improving the performance of seizure classification. Deep neural networks have the benefit of direct learning from data without the need for prior domain knowledge thus facilitating in classifying EEG signals without the presence of a feature set [47,48]. Medvedev et al. [20] in their paper, used LSTM for identifying the presence of spikes, ripples, and ripples-on-spikes. LSTM has been used by [49], to find the correlation between EEG samples such that it can be further used for seizure detection. The abnormality of neural activity during epileptic seizure attacks has been studied with the help of a multi-view deep learning model.

The current seizure detection methods discussed above are based on deep learning algorithms which attained promising results, but as seen most of the methods primarily rely on the CNN architecture. In few works, features are extracted using convolutional neural network and then used in traditional classifiers for classification. Generally, CNN use spatial information from EEG signals for detecting seizure dynamics. As EEG signals are principally nonlinear, hence isolating and extracting temporal features of brain activities using CNN may face some limitations [32]. In some literature wavelet transform and short-time Fourier transformation has been used to compute the time–frequency coefficients of EEG signals and also to transform signals into gray images [37]. These outputs are then used as input in 2D convolutional neural network for excerpting spectral and spatial information pertinent to epileptic seizures. The shortcoming of such approach is that one dimensional EEG signals are converted into two dimensional images or matrices which incur additional processing time and also increasing the volume of data. Another drawback of the existing methods is most of the methods are channel specific i.e. first it selects the channels and then use these selected channels features as input to detect seizure. Hence a method is proposed to avoid channel selection and use all input channel features to design a generalized epileptic seizure detection Also, commendable number of studies are dedicated towards patient-specific seizure detection method.

To address these limitations, the study proposes a novel deep learning method based on long short-term memory that is competent in designing an efficient automated seizure detection model from raw EEG signals with a simple architecture that is capable of consolidating all steps in a single system. The proposed long short-term memory network is such designed that there is no loss of generality even when there is a change in the EEG signal over some time. Despite scalp EEG being sensitive to artifacts, the proposed method has been able to reduce the false detection rate significantly and enhance the overall seizure detection accuracy of the system. When associated with other deep learning-based methods [32,33,39,40,43] the proposed single layer LSTM module has been effective in analyzing the hidden intrinsic temporal dependencies in EEG signals. Also, the proposed model used one-dimensional raw EEG data and did not require any transformation thus directly reducing the computational complexity of the model to a great extent. Furthermore, the proposed method did not need any handcrafted feature extraction technique as required in traditional classifiers, thus reducing substantial amount of preprocessing associated with feature engineering of EEG recordings. The proposed method has enhanced the seizure detection accuracy while following a generalized approach on the non-invasive CHB-MIT scalp dataset as well as for the invasive intracranial SWEC iEEG dataset hence, invoking the prospect of real-time implementation with high dimensional data.

The rest of the paper is arranged as follows. Section 2 briefly describes the methods used along with the architecture of the proposed approach and descriptions of the two datasets used. It also highlights the training and testing strategy of the work, the different learning parameters associated, and the performance measures for analyzing the results. The proposed model has been validated and tested on two publicly available datasets and the results obtained are mentioned in Section 3. Discussion of the achieved results with previous work is provided in Section 4. Finally, Section 5 concludes with the findings and highlights of the work.

2. Methods

2.1. Dataset description

2.1.1. Children's Hospital Boston Massachusetts (CHB-MIT)

The data set considered in this work consisted of scalp EEG recordings of pediatric patients between the ages of 1.5 to 15years suffering from intractable seizures. The dataset belongs to patients from the Children's Hospital Boston Massachusetts (CHB-MIT) [52]. The dataset comprising of 24 patients have been used for substantiating the proposed method. Each file contained one to a maximum of four seizures that were recorded using the bipolar placement of electrodes according to the 10–20 international electrode placement and were sampled at 256 Hz at 16-bit resolution. The number of EEG channels considered to be used as input in this work is 23. Fig. 1 shows the scalp EEG recoding of a single patient.

2.1.2. Sleep-Wake-Epilepsy-Center (SWEC)

A second dataset has been used for the validation of the proposed method and is taken from the Sleep-Wake-Epilepsy-Center (SWEC) of the University Department of Neurology at the Inselspital Bern and the Integrated Systems Laboratory of the ETH Zurich [53]. The database consisted of a total of 16 patients having 99 recordings with 100 seizures. The intracranial EEG (iEEG) signals have been recorded using the strip, depth, and grid electrodes. Localized between the 10/20 position of Fz and Cz, an extracranial electrode has been used as a reference for the recording. The sampling frequency of the signals is 512 Hz and a fourth-order Butterworth filter between 0.5 Hz and 150 Hz has been used to filter the recorded signals. Each of the EEG recordings comprised of interictal segments of 3 min followed by ictal segments ranging between 10 s and 1002 s and postictal segments of 3 min. Fig. 2 shows the intracranial EEG recoding of a single patient.

2.2. Classification methods

Different machine learning-based classifiers have been used for classifying between non-seizure and seizure EEG signals. In this work, three classifiers have been used, one popularly used traditional classifier and two deep learning-based classifiers. A brief description is mentioned of each method and detailed discussion of the parameters are stated in the proposed architecture.

2.2.1. Artificial neural network (ANN)

The human nervous system inspired the rudimentary architecture of the artificial neuron network (ANN). It is a data-driven model that is built on the foundation of mathematical models and

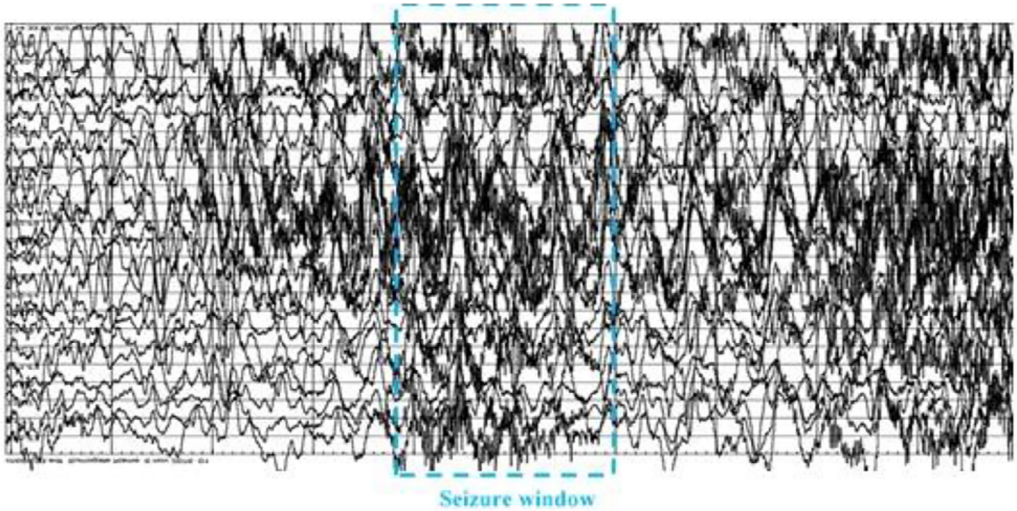

Fig. 1. Scalp EEG recording of a patient from the CHB-MIT dataset.

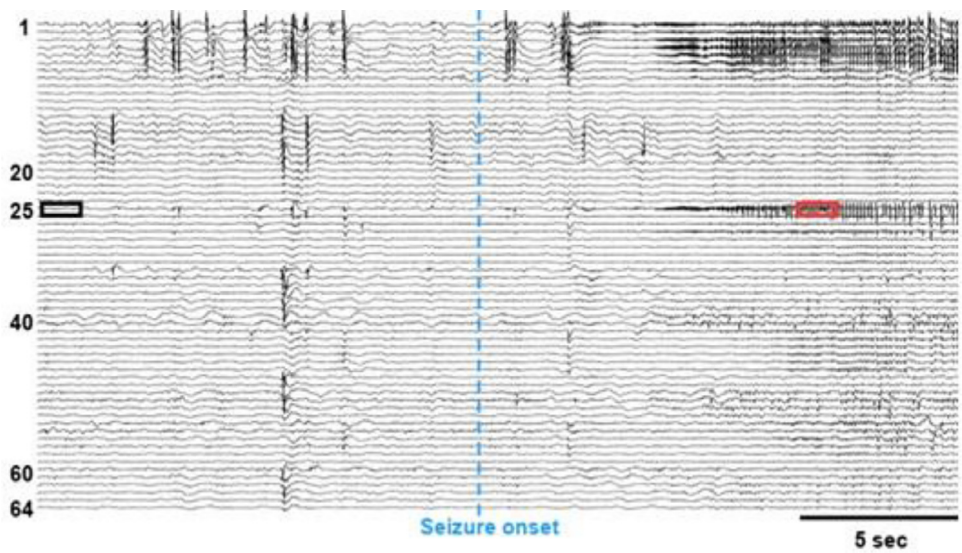

Fig. 2. Intracranial EEG recording of a patient from the SWEC dataset [54].

has the capability of recognizing intricate non-linear relations between inputs and outputs [25]. The base architecture of an ANN comprises an input layer followed by a hidden layer and an output layer. More than one hidden layer can be present in between the input and output layers that are linked with bias, weights, and different activation functions [34]. Characteristically, stochastic gradient descent optimization is used for training the neural network, and the backpropagation approach is used to update the weights. As loss function has significant relevance to the activation function, diverse loss functions are used depending on the type of problem [51]. In backpropagation,the error is propagated backward and thus tends to get reduced with each passing layer. This results in a longer training process with a gradual deterioration in accuracy and leads to the root cause of the vanishing gradient problem, a prominent drawback of the artificial neural network. Fig. 3, shows the network structure of the ANN used for epileptic seizure detection.

2.2.2. Convolutional neural network (CNN)

Convolutional neural network (CNN) is currently one of the preferred deep learning methods that gained importance in the last few years [42]. Although CNN has been primarily used in applications related to image processing, with advances it has been implemented for detection of medical signals using one and two-dimensional architectures [34]. 1D convolutional neural networks inherently are appropriate for processing biomedical signals like EEG recordings. The core architecture of the convolutional neural network is altered such that it is proficient in processing 1D EEG signals. Also, the 1D architecture is faster when compared to 2D structure, as it is straightforward and the number of parameters associated is less [24]. To learn the temporal, spectral, and spatial features from the raw signals, convolution layers are used in the network structure. The pooling layer reduces the dimension of extracted features by eliminating redundancy. The pooling layer also serves the purpose of avoiding overfitting of data. Even though there are different pooling methods, max-pooling is commonly used. The fully connected layer is a critical part of the convolutional neural network [50]. It performs the high-level reasoning needed for classification and the output of the fully connected layer provides the probability of each class label. Convolutional operation for 1-dimension is represented as:

$$x_j^L = f_a\left(\sum_{i=1}^{C} x_i^{L-1} \cdot ke_{ij}^L + bi_j^L\right) \tag{1}$$

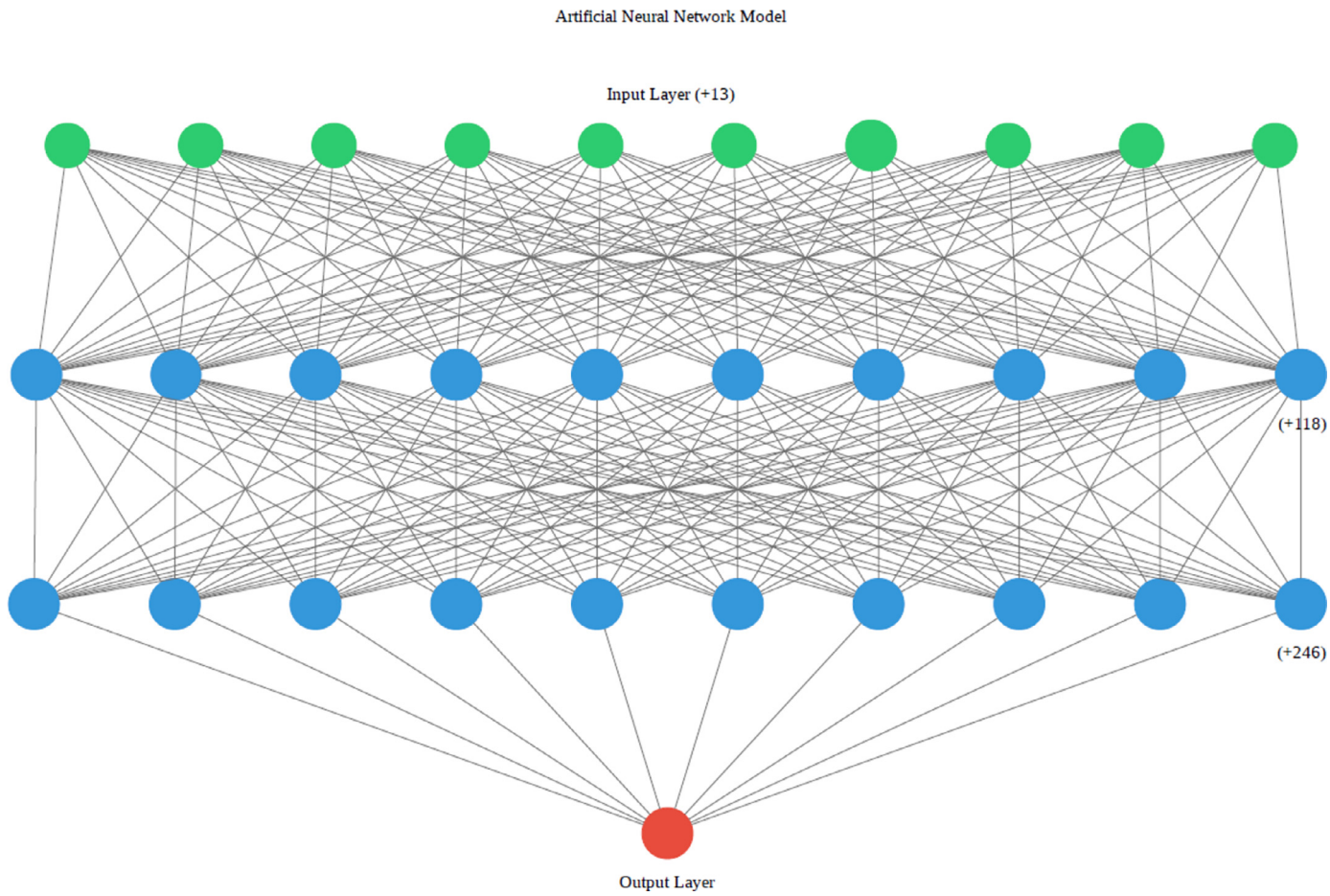

Fig. 3. Proposed structure of the artificial neural network.

where, L is the convolutional layer, f_a denotes the activation function, bi is the bias, ke represents the number of convolutional kernels and C stands for the number of input channels.

Batch normalization is given by the following equation:

$$BCN(x_{ke}) = \alpha_{MB}\left(\frac{x_{ke} - \mu_{MB}}{\sqrt{\sigma_{MB^2} + \varepsilon}}\right) \tag{2}$$

where,α_{MB} is the mini-batch variance, $\mu_{MB}$is the mean of mini-batch, σ_{MB^2} is a scalar parameter and ε represents random noise.

Adam optimizer uses the following equations for computing the decaying averages of past and past squared gradients:

$$mo_t = \beta_1 mo_{t-1} + (1 - \beta_1) gr_t \tag{3}$$

$$va_t = \beta_2 va_{t-1} + (1 - \beta_2) gr_t^2 \tag{4}$$

where, mo_t is the mean (first moment), va_t is the uncentered variance (second moment), β_1 and β_2 denote the decay rates and gr represents the gradient. Bias corrected first and second moment is given by:

$$\widehat{mo}_t = \frac{mo_t}{1 - \beta_1^t} \tag{5}$$

$$\widehat{va}_t = \frac{va_t}{1 - \beta_2^t} \tag{6}$$

Adam update rule is represented as:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\widehat{va}_t} + \varepsilon}\widehat{mo}_t \tag{7}$$

where, η is the general learning rate.

The loss function is calculated using binary cross-entropy for binary classification problem. The loss is obtained using the following equation:

$$Loss = -\frac{1}{output_size}\sum_{i=0}^{output_size} y_i \cdot log(\widehat{y}_i) + (1 - y_i) \cdot log(1 - \widehat{y}_i) \tag{8}$$

2.2.3. Long short-term memory (LSTM)

Long short-term memory (LSTM) is an advancement of the recurrent neural network (RNN) [24] that addressed the challenges of vanishing gradient and exploding gradient problems with the addition of more interactions per cell (or module). This special type of RNN has the capability of processing long-term dependencies and retaining information for an extended period. The LSTM model has a chain-like architecture with a different structure for the repeating module [20]. Unlike a typical RNN, the four interrelating layers have an exclusive technique of communication. Memory blocks known as cells are the inherent component of a typical LSTM [23]. Cell state is responsible for the unchanged forward flow of data [47]. Different gates having individual weighted values are present that are analogous to series or layer of matrix operations. These gates are responsible for the memorizing process and help in evading the long-term dependency problem. Generally, an LSTM architecture consists of the following gates – forget gate (Fg), input gate (x), cell candidate (n) and output gate (Op) [34]. The following equations represent the different gates.

$$Fg_t = \sigma\left(W_{Fg}[Hi_{t-1}, I_t] + B_{Fg}\right) \tag{9}$$

$$x_t = \sigma\left(W_x[Hi_{t-1}, I_t] + B_x\right) \tag{10}$$

$$n_t = tanh(W_n[Hi_{t-1}, I_t] + B_n) \tag{11}$$

$$Ce_t = Ce_{t-1}Fg_t + n_t x_t \tag{12}$$

$$\begin{aligned} Ce_t = {} & Ce_{t-1}\sigma\left(W_{Fg}[Hi_{t-1}, I_t] + B_{Fg}\right) \\ & + tanh(W_n[Hi_{t-1}, I_t] + B_n)\sigma\left(W_x[Hi_{t-1}, I_t] + B_x\right) \end{aligned} \tag{13}$$

$$Op_t = \sigma\left(W_{Op}[Hi_{t-1}, I_t] + B_{Op}\right) \tag{14}$$

$$Hi_t = Op_t tanh(Ce_t) \tag{15}$$

$$Hi_t = \sigma\left(W_{Op}[Hi_{t-1}, I_t] + B_{Op}\right) tanh(Ce_{t-1}Fg_t + n_t x_t) \tag{16}$$

$$\begin{aligned} Hi_t = {} & \sigma\left(W_{Op}[Hi_{t-1}, I_t] + B_{Op}\right) tanh\Big(Ce_{t-1}\sigma\left(W_{Fg}[Hi_{t-1}, I_t] + B_{Fg}\right) \\ & + tanh(W_n[Hi_{t-1}, I_t] + B_n)\sigma\left(W_x[Hi_{t-1}, I_t] + B_x\right)\Big) \end{aligned} \tag{17}$$

2.3. Architecture of the proposed method

The schematic diagram of the proposed method is presented in Fig. 7. In this work, raw EEG signals have been used for the detection of epileptic seizures. As a basic butter-worth filter is used during the recording of the EEG signals, additional pre-processing of the recordings is not performed. As the aim of the work is to develop channel independent seizure detection framework, all the recorded 23 channels of EEG signals are used as input. The proposed method follows a generalized approach and the entire dataset is randomly divided into two subsets – training and testing. A further split of the training data is done in the ratio of 80:20 to perform simultaneous validation of the trained module. This is followed by the selection of the classifier model along with the learnable parameters. Both traditional and deep learning models have been used for the study of epileptic seizures. The base network structures are placed in the collection module. During the initiation phase, the network structure, layers, different learnable parameters of the optimizer, and loss function are initialized for the training phase. This is followed by the interpretation and comparison of the model based on their training time and selection of the hyper-parameters. The cross-validated trained models are kept for future interpretation of other datasets. Apart from the evaluation, the trained models are further tested on the separate test dataset that is partitioned at the beginning of the proposed approach. The proposed method helps in understanding and analyzing the efficiency of each model, the loss rate, and convergence of the model with respect to epochs.

The network structure of the artificial neural network as seen in Fig. 3 consisted of an input layer with 128 neurons followed by a hidden layer of 256 neurons and a final output layer. Both the input and the hidden layer have ReLU as the activation function and the output layer has sigmoid as the activation function. Fig. 4 shows the proposed architecture of the 1D convolutional neural network that is capable of feature extraction and classification. The convolutional neural network has two 1DConv layers that are used to learn the features from the EEG signals. Each convolutional block consisted of a convolution layer with 128 neurons with a non-linear activation function – rectified linear unit (ReLU). The third layer is batch normalization which handles the internal variable shift. The final layer in the convolutional block is the max-pooling layer which presents the translation invariance. The second convolutional block has the same structure as the first block. The feature maps obtained from the convolutional layer are flattened into a one-dimensional vector by the flatten layer and further linked to the fully connected dense layer for incorporation and classification of features. The first three dense layers have tanh as the activation function and are followed by a dropout layer. The final dense layer has the sigmoid activation function that will generate the output vector according to the two categories. The other deep learning model used is long short-term memory and the proposed structure of the LSTM network is presented in Figs. 5 and 6 shows a single LSTM cell structure. The LSTM block consists of 256 memory units with ReLU as the activation function. These units are made up of different gates. The forget gate removes unwanted information not required by the LSTM and optimizes the performance of the LSTM network. The addition of information is performed by the input gate using the tanh function while the regulatory filter is multiplied using the sigmoid function and is added to the cell state. Extraction of useful information is achieved by the output gate. The LSTM unit is followed by two fully connected dense layers. The first dense layer has 128 units and a non-linear activation function (ReLU), while the final layer has the sigmoid activation function. The detailed description of all the classifiers is mentioned in Table 1.

2.4. Learning parameters of different models

The weight parameters associated with each of the network structures are required to be learned from the dataset during the training phase of the proposed method. In the artificial neural network, binary cross-entropy is used as the loss function. Adam optimizer is used and different hyper parameters of Adam algorithm used are, decay rate ($beta_1 = 0.9$), square gradient decay rate ($beta_2 = 0.999$), and learning rate (0.0001). The batch size chosen is 200 with 100 training epochs. The parameters for the convolutional neural network are also learned using the conservative approach of back propagation and binary cross-entropy is used as the loss function. RMS Prop and stochastic gradient descent have been used together as Adam optimizer that is created on the adaptive estimation of the first-order moment (mean) and second-order moment (uncentered variance). The different hyper parameters of Adam algorithm used are, decay rate ($beta_1 = 0.9$), square gradient decay rate ($beta_2 = 0.999$), and learning rate (0.001). The batch size chosen for training the 1D CNN is 600 with 200 training epochs. Similarly, for the long short-term memory network, the loss function is kept the same as binary cross-entropy along with similar hyper parameters of the Adam optimizer. LSTM has the tendency of over fitting training data, thus impacting the predictability of the network. To avoid over fitting, L_2 regularization has been used for regularizing the weight of the nodes in the LSTM network. The weight decay or L_2 regularization factor used for training the model is 0.0001. The batch size used for training the LSTM model is 600 with 1000 training epochs and a sequence length of 5.

2.5. Training and testing strategy

The performance of the proposed method has been assessed using cross-validation to check the reliability of the proposed results. The EEG signals have been divided into N mutually exclusive folds that have an equal span of non-seizure and seizure recordings. Out of the N folds of data, one-fold has been kept aside as test data to evaluate the performance of the trained models. Therefore, $(N - 1)$ folds have been used for training and validating the models. To evade the over fitting problem, in each of the fold validation split has been done such that 20% of the training data is

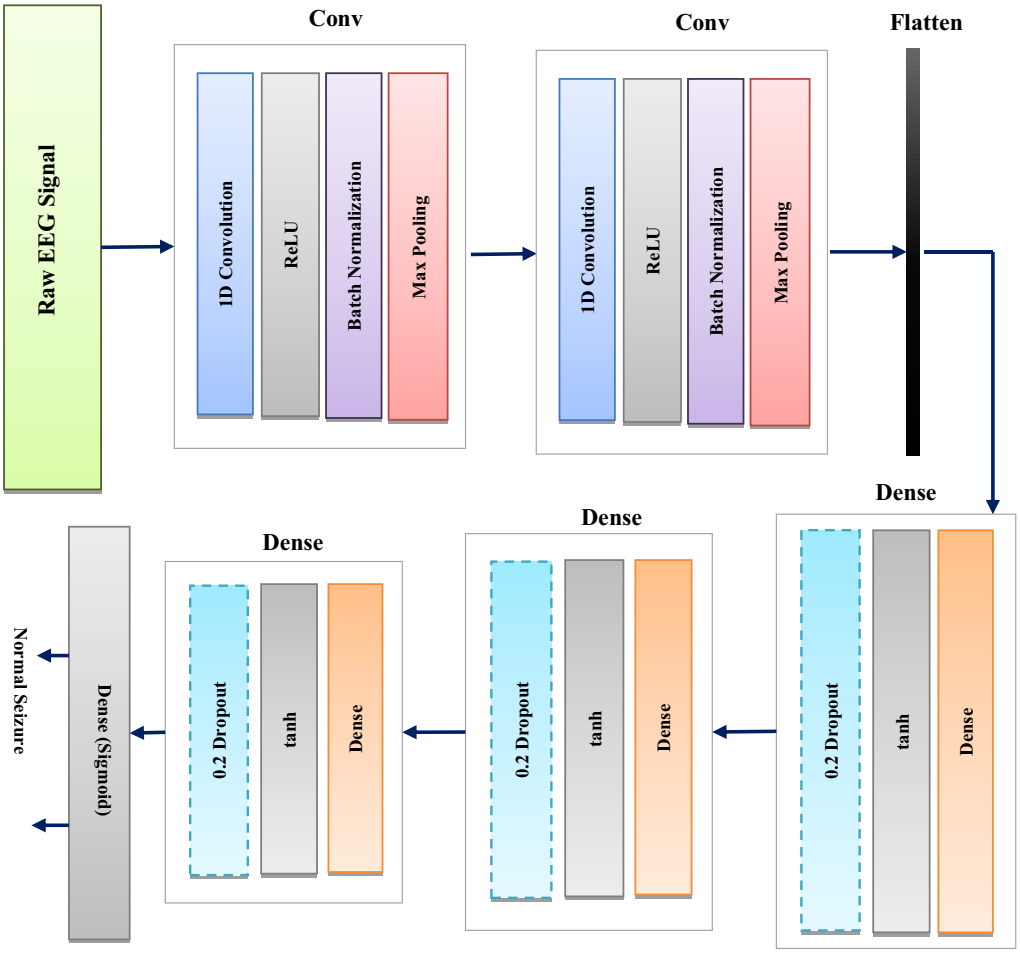

Fig. 4. Proposed 1D convolutional neural network architecture.

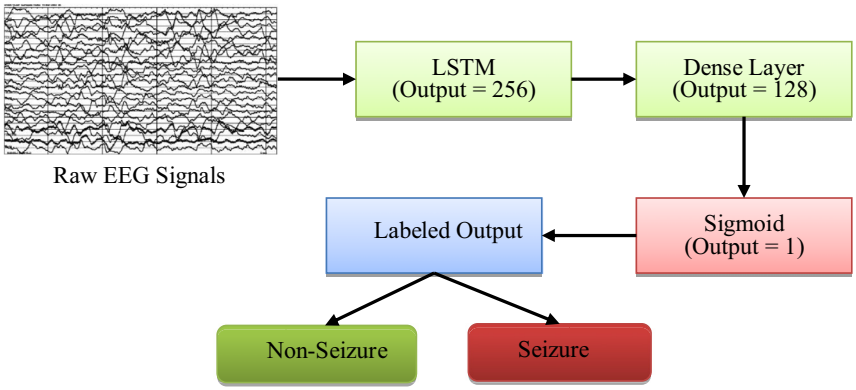

Fig. 5. Proposed long short-term memory architecture.

used for simultaneous validation of the model while the model is trained with the residual 80% of data. With the completion of each epoch, the error rate is calculated so that the model does not over fit the training data. Generally, if the error rate keeps on increasing continuously for four successive epochs then the training process is terminated. After the models are trained, the test fold is used for evaluating the training model and this process is repeated N times unless all the folds are exhausted. The average of all the N individual measures gives the cross-validation result. In this work, 10-fold cross-validation has been used for analyzing the performance of the proposed method.

2.6. Performance measure

In this work, sensitivity, specificity, accuracy, Matthew's correlation coefficient, and F_1 score have been used to study the performance of both traditional classifier and deep learning methods. Matthew's Correlation Coefficient (MCC) is a special case and equivalent to Karl Pearson's phi coefficient. It helps in measuring and understanding the quality of a given classifier. Positive and negative of both true and false cases are taken into account when the coefficient is calculated and therefore this measure is considered a balanced measure. The coefficient value lies between −1

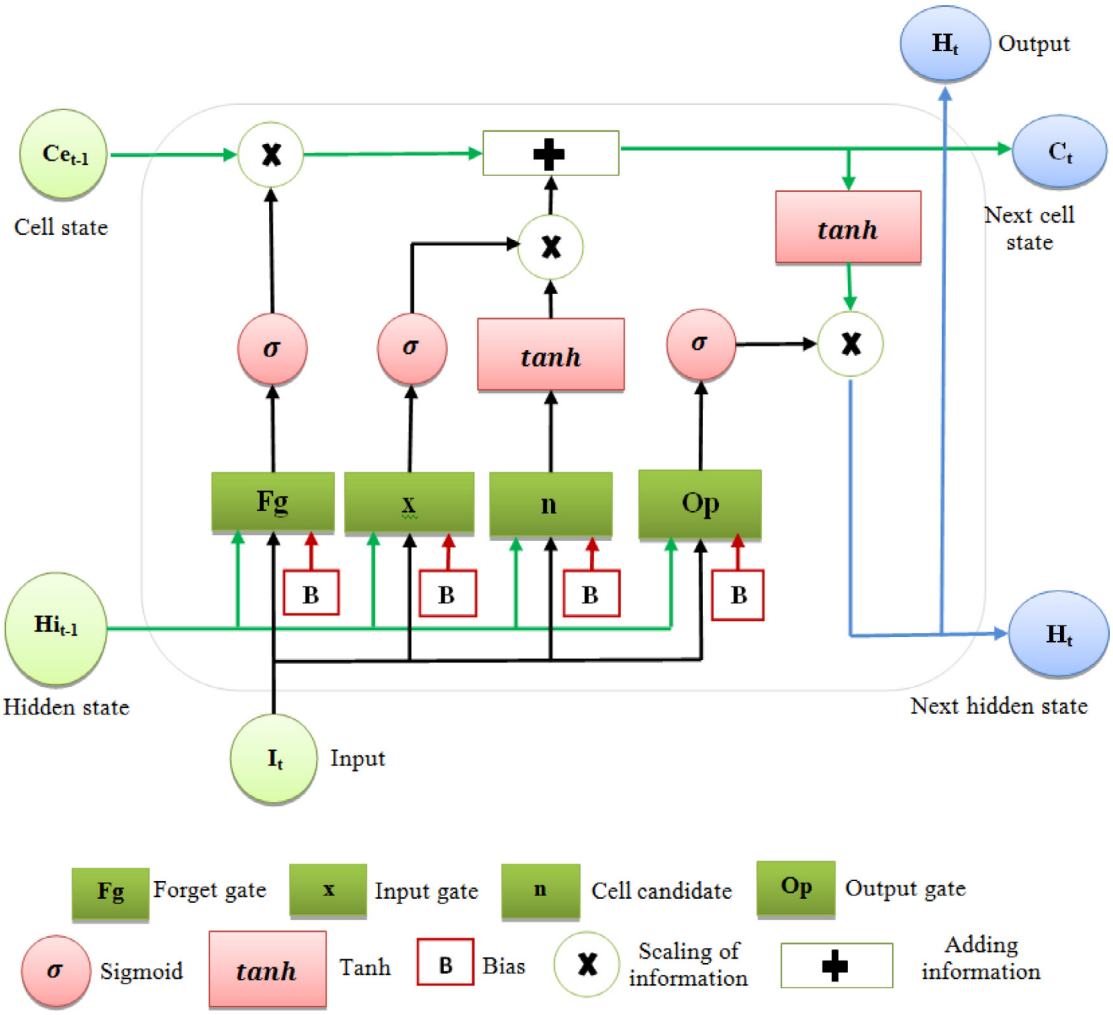

Fig. 6. Different gates in a LSTM cell.

Table 1
Description of the classifiers.

Classifier	Block	Type of layer	Number of neurons	Other parameters
Artificial neural network (ANN)	Input	Dense	128	Activation: ReLU
	Hidden	Dense	256	Activation: ReLU
	Output	Dense	1	Activation: sigmoid
1D Convolutional Neural network (1D-CNN)	Conv1	Convolution	128	Kernel size: 3
		Activation	128	Activation: ReLU
		Batch normalization	128	-
		Max pooling	128	Pool size: 2
	Conv1	Convolution	128	Kernel size: 3
		Activation	128	Activation: ReLU
		Batch normalization	128	-
		Max pooling	128	Pool size: 2
	FC 1	Dense	64	-
		Activation	64	Activation: tanh
		Dropout	64	0.2
	FC 2	Dense	32	-
		Activation	32	Activation: tanh
		Dropout	32	0.2
	FC 3	Dense	64	
		Activation	64	Activation: tanh
		Dropout	64	0.2
	FC 4	Dense	1	
		Activation	1	Activation: sigmoid Regularization: L_2
Long short-term memory (LSTM)	LSTM	LSTM	256	Activation: ReLU
	FC	Dense	128	Activation: ReLU
	FC	Dense	1	Activation: sigmoid Regularization: L_2

*FC-Fully connected.

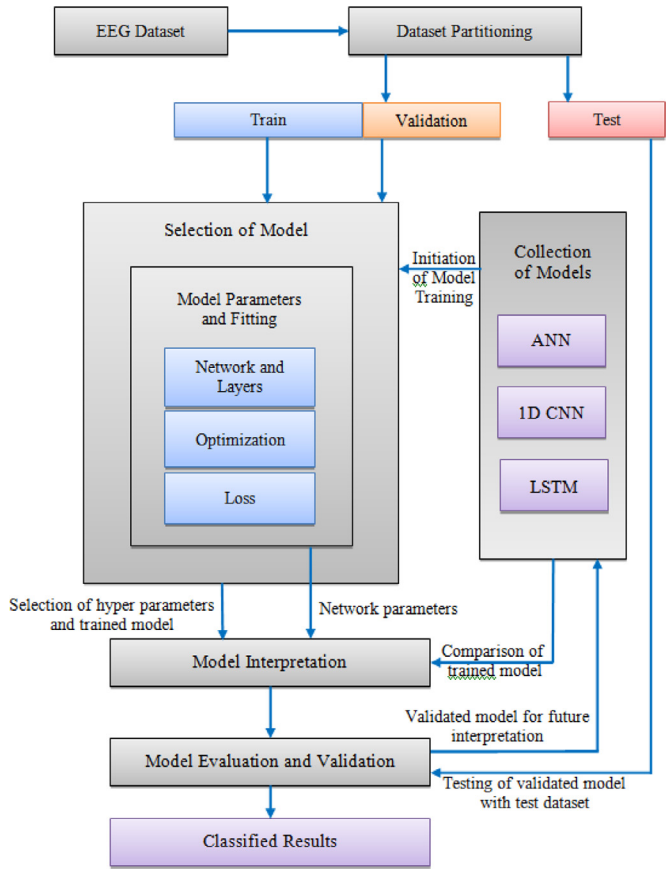

Fig.7. Schematic diagram of the proposed deep learning method.

and +1. MCC is expressed as,

$$MCC = {TP \times TN - FP \times FN}/{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \tag{18}$$

where, TP is true positive, TN is true negative, FP and FN represent false positive and false negative events, respectively.

F_1 score takes into account both precision and recall to calculate the F–score. It is also a statistical analysis of the classification. F_1 score or F-measure represents the harmonic mean between precision and recall. It is sometimes known as the Dice similarity coefficient.

$$F_1 = 2 \cdot \frac{precision \cdot recall}{precision + recall} \tag{19}$$

where, $precision(PPV) = \frac{TP}{TP+FP}$.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{20}$$

$$Sensitivity(Recall) = \frac{TP}{TP+FN} \tag{21}$$

$$Specificity = \frac{TN}{TN+FP} \tag{22}$$

3. Results

Traditional as well as deep learning method has been validated and tested with the proposed method on raw EEG signals from two publicly available datasets. The performance of different classifiers has been assessed using metrics such as accuracy, sensitivity, specificity, precision, false-positive rate, Matthew's correlation coefficient (MCC), and F1 score. Convergence analysis has also been done to understand the correctness of the proposed method.

3.1. Seizure detection performance of different classifiers on CHB-MIT dataset

In the present study, artificial neural network (ANN), 1D convolutional network (1D-CNN), and long short-term memory (LSTM) have been used for designing an automated epileptic seizure detection system. All the three classifiers have been used on two separate EEG datasets consisting of seizure and non-seizure recordings. For all the approaches, 10-fold cross-validation has been used, and $(N - 1)$ folds of training data are further split for validation. The trained module is then tested with the fold that has been kept aside for testing. Results of the binary classification on the first dataset (CHB-MIT) are presented in Table 2. The deep learning methods gave better classification accuracy than the traditional approach for raw EEG signals. The false positive rate is 0.003/h for LSTM while 1D-CNN and ANN detected the presence of seizure with a false-positive rate of 0.034/h and 0.044/h ,respectively. As the rate of false detection is less, the precision of the LSTM network is also higher among the three classifiers. Fig. 8 shows the accuracy curve of two methods with respect to training epochs. The plot for LSTM as seen in Fig. 9 is slightly different as it shows both accuracy and loss of the module during validation.

3.2. Seizure detection performance of different classifiers on SWEC dataset

In this work, a second dataset has been used to test the efficiency and capability of the proposed method. The dataset used belonged to the Sleep-Wake-Epilepsy-Center (SWEC) and contained intracranial EEG (iEEG). The model configuration and parameters of the respective classifiers remained unchanged when the second dataset is validated. The seizure detection performance on the SWEC dataset is presented in Table 3. As seen from the results, the performance metrics were similar to the CHB-MIT dataset. The trait of an effective seizure detection system is maximizing the seizure detection sensitivity and minimizing the false detection rate. The proposed method has been able to achieve this criterion. The false-positive rate is 0.006/h for LSTM while 1D-CNN and ANN detected the presence of seizure with a false-positive rate of 0.002/h and 0.075/h, respectively which is analogous to the previous dataset. Fig. 12 presents the performance analysis of both the datasets based on F1 measure and false positive rate per hour (/h). Fig. 10 shows the accuracy and loss curve of the LSTM network when tested with the SWEC dataset.

3.3. Convergence analysis of the proposed method

Deep learning methods have the capability of performing gradient-based optimization and thus helps the algorithm to converge to the global optimum. An algorithm is said to converge to the lowest error when it generates an error in sequence for a period. This signifies the approximate balance and correctness of the algorithm. In this work, the convergence of the loss function has been studied to analyze the convergence of the proposed method. Fig. 11 represents the convergence curve of the loss function. It can be observed, the objective function remains the same for several epochs. The loss function for the LSTM network decreases gradually and becomes stable after certain iteration. The convergence for the second dataset (SWEC) can be analyzed from Fig. 10. The loss function converges and remains unchanged over a prolonged period.

Table 2
Performance of different learning model using 10-fold cross-validation on CHB-MIT dataset.

Classifier	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	False positive rate (per hr.)	F_1 Score	MCC
ANN	94.3	92.3	95.8	95.6	0.044	0.940	0.872
1D-CNN	96.5	94.7	97.1	96.9	0.034	0.958	0.899
LSTM	99.9	99.9	99.9	99.9	0.003	0.999	0.999

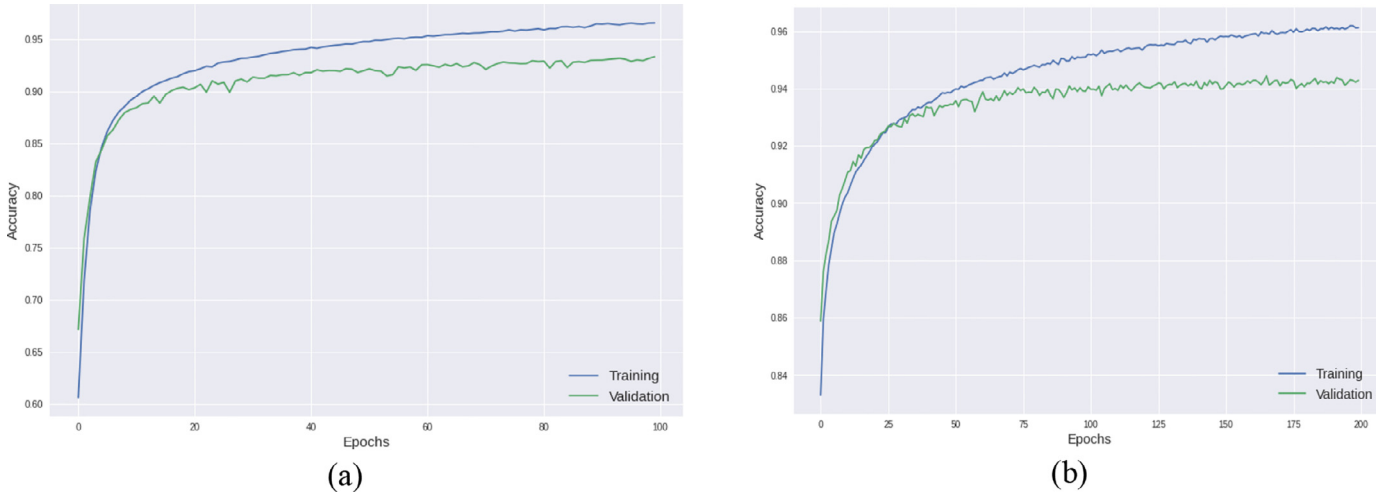

Fig. 8. The accuracy vs epochs plot of (a) artificial neural network, (b) 1D convolutional neural network for CHB-MIT dataset.

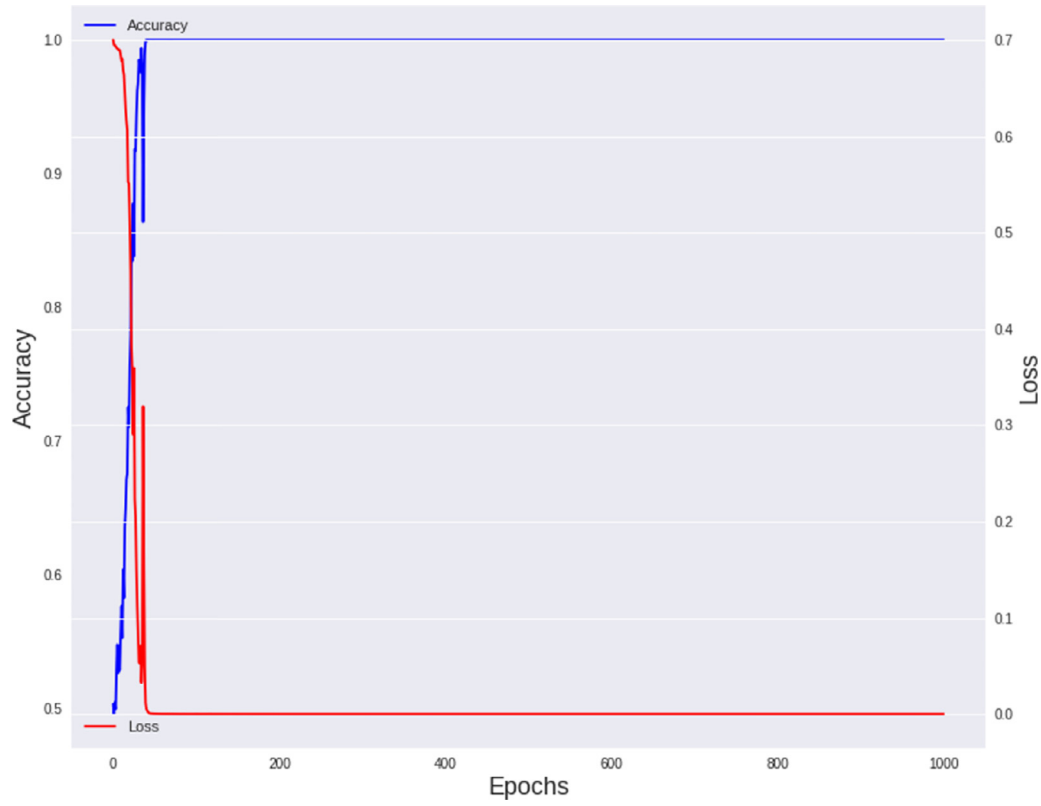

Fig. 9. Accuracy-loss vs epochs plot of long short-term memory network for CHB-MIT dataset.

Table 3
Performance of different learning model using 10-fold cross-validation on SWEC dataset.

Classifier	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	False positive rate (per hr.)	F_1 Score	MCC
ANN	94.7	94.4	95.6	95.4	0.075	0.938	0.880
1D-CNN	98.1	97.3	100.0	100.0	0.002	0.986	0.974
LSTM	99.5	99.4	99.3	99.2	0.006	0.993	0.987

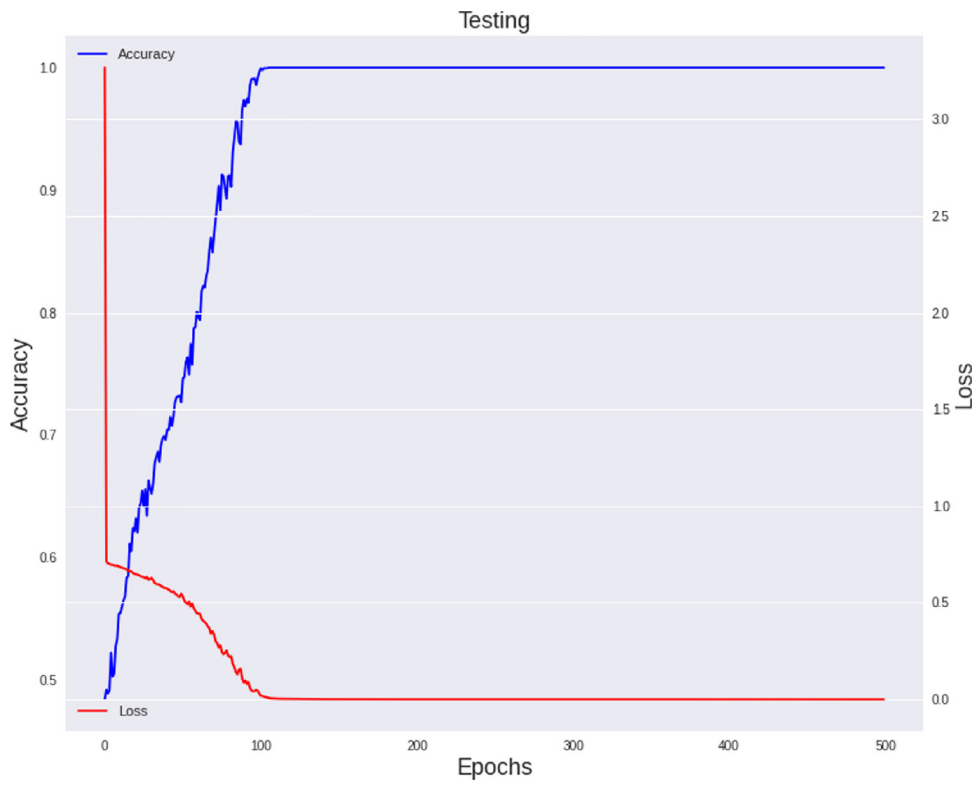

Fig. 10. Accuracy-loss vs epochs plot of long short-term memory network for SWEC dataset.

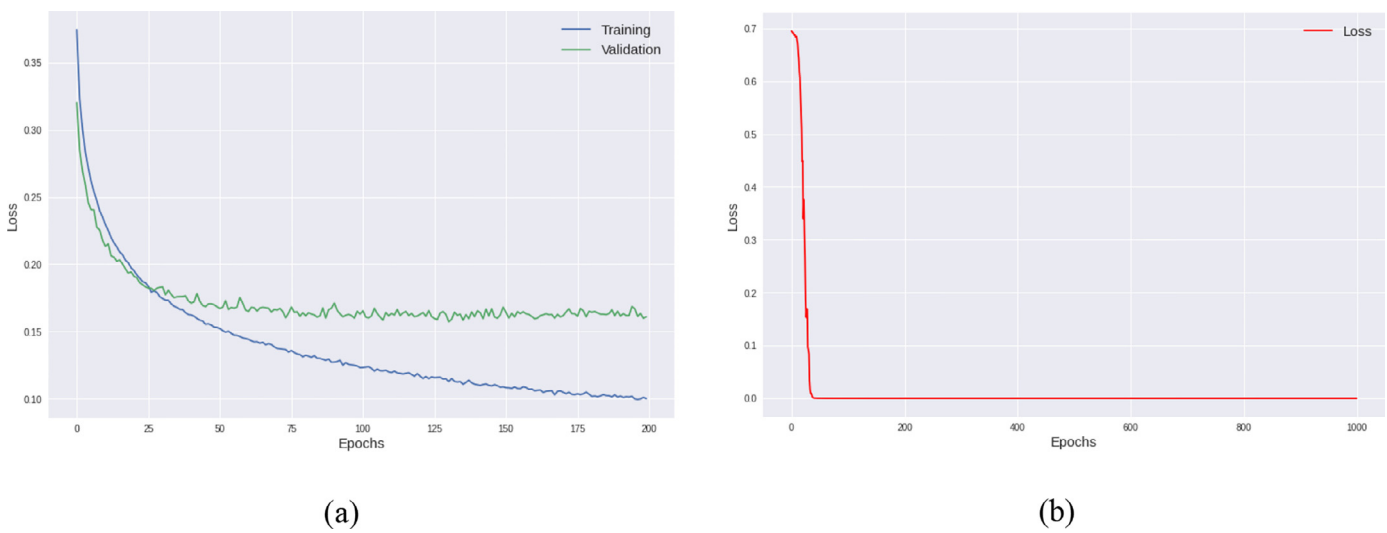

Fig. 11. Convergence of loss value of (a) 1D-CNN and (b) LSTM for CHB-MIT dataset.

4. Discussion

The proposed method is pitched forward as a seizure detection system for both non-invasive (scalp) as well as invasive (intracranial) EEG recordings. The proposed method is channel independent as it does not select any EEG channel but takes the entire recorded channel as input. This section compares the performance of the proposed method with other related methods mentioned in the literature such that having used the same dataset for analyzing the detection capability of the system. Table 4 highlights the comparison of the present proposed approach with previous works. Different seizure detection methods have been studied where either non-invasive or invasive EEG recordings are used. The use of both types of recording in the same module for seizure detection is fewer in number. The proposed work uses non-invasive as well as invasive EEG signals thus aiming for a robust and comprehensive system that would seamlessly adapt and work efficiently in detecting the presence of a seizure segment.

Sensitivity is one of the major aspects of an automated seizure detection unit. The recall signifies the rate of correct identification of seizure events from an EEG recording. The proposed LSTM model has been able to achieve a higher sensitivity rate using a generalized approach, while Tsiouris et al. [32] attained the same with the patient-specific method. The authors have gradually incremented the segmentation of EEG recordings using the concept of a time window and that has impacted the sensitivity to an ex-

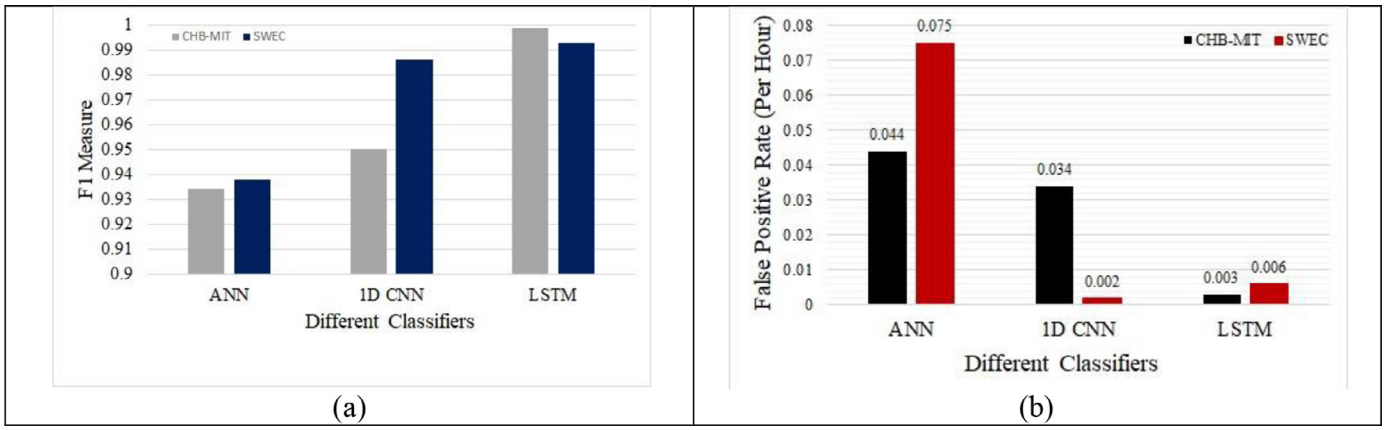

Fig. 12. Performance analysis between the two datasets based on (a) F1 measure and (b) false positive rate (/hr.).

Table 4
Comparison between the proposed method and previous works on the CHB-MIT dataset.

Authors	Methods	Validation method	Accuracy (%)	Sensitivity (%)	Specificity (%)	FPR/h	F_1 Score
Boonyakitanont et al. [25]	ScoreNet, CNN, SVM	LOOCV	99.8	76.5	99.2	0.09	0.701
Hamavar and Asl [26]	Change point Detector	-	-	100.0	-	0.39	-
Yuan et al. [29]	STFT-SSDA	Cross patient	93.8	-	-	-	0.960
Hossain et al. [33]	Deep CNN	Cross patient	98.05	90.0	91.65	-	-
Wei et al. [36]	CNN, Data augmentation	Cross patient	84.0	72.1	95.8	-	-
Li et al. [30]	FC-NLSTM	10-fold CV	95.29	95.42	95.29	0.66	-
Gómez et al. [43]	CNN	LOO	99.3	58.3	99.6	<0.5	0.59
Wen and Zhang[27]	AE-CDNN	5-fold CV	88.1	-	-	-	-
Cao et al. [37]	STFT, CNN	-	90.13	-	-	0.07	-
Zhou et al. [40]	CNN	-	95.6	94.2	96.9	-	-
Usman et al. [38]	CNN, SVM	-	-	92.7	90.8	-	-
Hu et al. [39]	Bi-LSTM	-	-	93.6	91.8	-	-
Tsiouris et al. [32]	LSTM	-	-	99.8	99.8	0.02	
This work	LSTM	10-fold CV	99.9	99.9	99.9	0.003	0.999

CV- Cross validation; SSDA- Stacked sparse denoising autoencoders; LOOCV- Leave-one-out cross validation; CNN- Convolutional neural network; Bi-LSTM- Bi-directional long short-term memory; FC- Fully connected; NLSTM- Nested LSTM; STFT- Short-time Fourier transform; AE-CDNN- Autoencoders based deep convolutional neural network.

tent. The proposed LSTM model was capable of performing better than Bi-LSTM [38] that has the advantage of using information back and forth, in terms of sensitivity. The generalized approach has been effective in adapting to the subtle changes in the signals and identifying maximum of the seizure period than the patient-specific strategy applied by Gomez et al. [43], making the work more suitable for real-life implementation.

Another important aspect of the proposed work lies in the false positive rate. As the identification of seizure is crucial, a minimal value of the false-positive rate is desired by almost all seizure detection units. The proposed method has been able to accomplish a low false positive rate and fared exceptionally when compared with other related work [25,26,30,32,37,43], hence solidifying the stability of the system. Therefore, the present work when compared in terms of FPR/h has the potential of being implemented in a real-life situation and would aid in the design of an alarm generation system. The proposed method has simple architecture unlike [39], where Bi-LSTM is used for extracting temporal features before further classification. In spite of applying nested LSTM [30], the false positive rate fared below the proposed approach.

Deep learning methods have the inherent quality of learning features from a given dataset. The proposed method has been designed keeping this aspect in mind. Therefore, the proposed LSTM network has been used single-handedly in analyzing the EEG recordings. Even though the method lacks intricate and deep network structures [27,33,36] but when accuracy is compared the proposed method has achieved higher accuracy in correctly identifying epileptic seizures. Hence, it is worth observing that high classification may be achieved due to different data processing techniques, design of model, and selection of parameters. But at times these can limit the performance of the system with slight changes in the constraint. Thus, epilepsy seizure detection system needs models that have autonomous learning ability and performance is not hindered by parameters, and architectural complexity.

Generally, studies with the classification of EEG recordings tend to use a single dataset for designing the proposed framework. But in this work, a scalp EEG dataset and an iEEG dataset have been used in a single framework while maintaining all the configurations. As iEEG tends to use a large number of channels, the proposed architecture has been able to work efficiently with the added number of EEG channels. The sequence length of the LSTM cell has been selected such there is no loss in generality. EEG signals tend to show subtle changes for a brief period and the proposed architecture is designed in a way that information loss can be avoided even when there is a change in data over time. Also, the proposed method avoids the time-consuming step of feature extraction and once a module is trained and stored in the model collection, it can be used in the future for detection and analysis of EEG recordings. This would eventually help in enhancing the quality of life of a patient suffering from intractable epilepsy.

The limitation of the work lies with the framework of the proposed method. As both the dataset have short-duration signals, the generalized trained module has been able to classify the seizures accurately. But with long-duration EEG recordings, if there is an imbalance in the preictal and interictal data, segmentation would be required such that deep learning architecture can learn the fea-

tures efficiently. Also, iEEG recordings vary in channels due to strip or grid placement of electrodes, hence, a generalized trained module is required that can accommodate and facilitate the varying number of channels. The present proposed method did not emphasize the detection latency of the seizure. This can be incorporated in future work as it will help in designing an effective alarm generation system that would initiate an alarm whenever changes are observed prior to seizure onset.

The most critical advancement in machine learning in last few years is the development of deep learning algorithms. 1D convolutional networks are useful in biomedical because of compact and simple configuration of the networks. Convolutional neural network is considered as black box as it has the inherent capability of learning features without the need of additional feature extraction as both feature extraction and classification steps are merged into one process. In this work, as raw EEG signals have been used, hence the 1D-CNN is used to extract optimized features that maximized accuracy and minimized the false detection rate. Generally, a CNN layer comprises of five operations: (i) Convolution with linear time invariant filters (ii) Summation of diverse features (iii) Calculation of local summaries (iv) Subsampling (v) Pointwise nonlinear activation function. The proposed CNN consisted of sequential layers composed of non-casual convolution, pooling and batch normalization. Signals from multiple EEG channels act as input to the first layer. Apart from processing of features using linear filters, summaries of feature components are made in the max-pooling layer. Also, as pooling layer is involved with subsampling of data, in this work a pool size of 2 has been used to reduce the dimensionality with minimum loss in important information. This is followed by the non-linear activation function that produces the features which becomes the input for the next layer. To avoid over fitting of data, regularization with dropout and batch normalization have been used to fine tune the network. The training of the network has been done multiple times before finding the suitable configuration that minimized the network loss and output difference.

The learning capability of the LSTM model is used to evaluate the implication of each feature in the final decision making, as there is no discrete feature selection step prior to classification. The feature space is internally evaluated for all patients by the LSTM model. It learns the most useful features and adopts a generalized approach in order to produce more effective, and robust seizure detector. The LSTM network in the proposed method received input from multiple EEG channels and the EEG recordings have been assigned to the network in sequence of preictal and ictal recording. The sequence of the LSTM network is altered and tested before settling the final sequence number. As generalized approach has been used by the LSTM network, large number of EEG segments comprising of both preictal and ictal phases have been included and this helped in distinguishing between the variations in EEG signals. To avoid over-fitting, the EEG segments for each patient are randomly selected and shuffled before being fitted as input into the LSTM network. EEG signals are complex and the relationship among the different domains (spatial, temporal and spectral) is strong. Therefore, when input data is fed into the model, keeping the relations intact is crucial [55]. In this work, as all the EEG channels have been considered therefore, column kernel has been used in the LSTM unit such that all the channels are covered with minimum distortion. The seizure detection ability of the proposed method has been evaluated using 10-fold cross-validation. As a generalized approach is considered, the LSTM network is able to learn a broad pattern of preictal and ictal segments pertaining to different types of seizures originating from various locations in the brain. In long-term EEG recordings, imbalance of data is a common issue as ictal segments are comparatively fewer than preictal segments. Therefore, to avoid this issue of class imbalance, the EEG segments are selected in a way that durations of preictal and ictal matched. In deep learning architecture, there are no specific directives regarding the width and depth of a deep learning model [15]. Generally, importance is given to learning capability of the model which should be in harmony with the classification problem while keeping the complexity of the model low to decrease the computational cost.

In various previous works, variants of LSTM models [30,32,46] have been used by authors. Some have used nested LSTM model while in other convolutional LSTM [44], deep-CLSTM [45] and bidirectional LSTM network [20] have been used for studying seizure, ripples, spikes and so on. When compared to other LSTM models used previously, the number of units in the LSTM layer is large and that has helped in efficiently distinguishing between normal and seizure activity from raw EEG signal. Thus, the proposed model has shown the capability of discerning epileptic signals without the need of an additional step of feature extraction while in other LSTM models, either feature are extracted or an additional convolutional layer is added to extract features before being fed to the LSTM network. As training a LSTM network is essentially time consuming, and EEG signals results in large dataset, hence a simple network has been selected that would bring down the training time comparatively. Also, when the sequence length of the signal is changed, the number of parameters generated remains the same, the network size is not affected which is another promising advantage of LSTM network. In this work, a single-layered LSTM model has been used that has different sequence length and parameters for detecting epileptic seizure. Although a simple LSTM framework has been used for the proposed method, the network structure is different. The number of units of the LSTM layer has been altered and modifications have been made and tested before finalizing the proposed network structure.

In this work, as all EEG channels have been considered, therefore seizure originating at different locations has been detected by the proposed method efficiently. Equal weightage has been given to each channel without any priority or selection. As seen during a partial seizure, few specific channels of a hemisphere reflect the abnormal electrical activity that accompanies a seizure attack. While in generalized seizure attack, multiple channels from both hemispheres of the brain are affected. As in this work each EEG recording channel is given equal importance, hence chances of missing seizure onset or presence of seizure is low. Due to this seizure initiating at any location can be detected by the proposed method. Also, the amount of time required by the proposed method is minimum as it can produce output as soon as the EEG samples are produced in time domain. LSTM networks are said to be suitable for time series signals and as EEG recordings belong to the same category, utilizing the capabilities of LSTM network has been done for seizure detection. The subtle changes present in the EEG recordings over a time frame can be analyzed using the facility of sequence present in the LSTM network. The competency of retaining information over a time period helps in understanding and analyzing the EEG signal. Looping through the time sequence helped in better training of the EEG signal which in turn helped in better seizure detection with maximum accuracy and minimum error.

5. Conclusion

In this work, a channel independent pediatric epileptic seizure detection framework has been proposed. All the recorded channels are used as input instead of selected channels so that generalized seizures will be easy to detect. EEG recordings are complex and generate a huge volume of data and to handle such a humongous amount of data, deep learning models have entered the research area revealing their capabilities and effectiveness.

These deep learning techniques have unlocked innovative chances in addressing different biomedical challenges, one of them being epileptic seizure detection. LSTM networks have been used in this work as the proposed method for analyzing and detecting epileptic seizures. The proposed approach has been able to provide a commendable false positive rate of 0.003/h with an F_1 score of 0.999. When the performance metrics are compared for the CHB-MIT, the proposed work recorded high sensitivity and low FPR which is generally considered ideal for the seizure detection system. Therefore, extensive testing is required with long-duration EEG recordings such that the method can be implemented in the clinical surrounding.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Declaration of Competing Interest

The authors declare that they have no conflict of interests. The authors alone are responsible for the content and writing of this article.

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